

Birla Institute of Technology & Science, Pilani
1st Semester 2019-2020
Comprehensive Exam-Part A (Set I) (Close Book)

Course No: EEE F311
Maximum Marks: 20

Course Title: Communication Systems
Duration: 60 Minutes

Date: 09-12-2019

Please write the final answer in the space provided.
Rough work should be done in the main answer script.

- Q1. A phase modulated (PM) signal is given as $x(t) = 20 \cos(200\pi t + 12\pi m(t))$ where the message signal $m(t) = \Delta(t/6)$ and Δ denotes the triangular signal. Find the peak frequency deviation (in Hz) if the carrier frequency is 100 Hz.

Ans:

2

- Q2. There are 7 alphabets with probabilities $\{0.05, 0.1, 0.15, 0.17, 0.18, 0.22, 0.13\}$. Using a ternary Huffman code, find the average length of code.

Ans:

1.78

- Q3. A generator polynomial $g(x) = x^2 + 1$ construct a $(4, 2)$ cyclic code. Find the error detecting capability of the code.

Ans:

1

- Q4. Write down two types of single-user spread spectrum techniques.

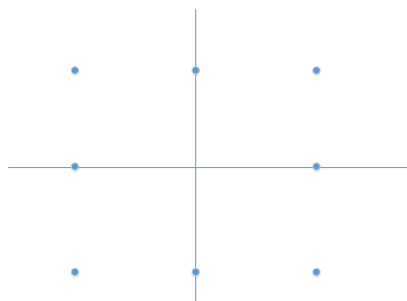
Ans:

Frequency Hopping [1 Mark] and Direct Sequence [1 Mark]

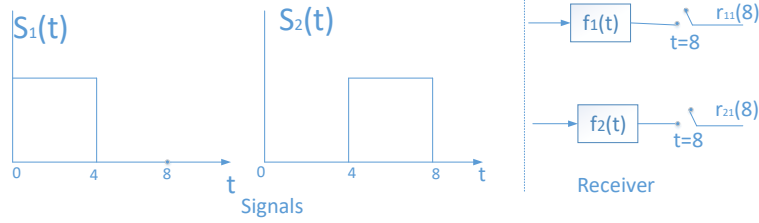
- Q5. Determine the average transmitted power for an 8-point QAM signal constellation as given in the figure below. The minimum distance between two adjacent constellation point is $d_{min} = 2$.

Ans:

Symbol power= 6 or Bit power= 2



- Q6. In a binary transmission rectangular pulses $s_1(t)$ and $s_2(t)$ are transmitted under noiseless channel condition. Receiver uses two matched filters $f_1(t)$ and $f_2(t)$ corresponding to $s_1(t)$ and $s_2(t)$. Assuming that $s_1(t)$ was transmitted, find $\frac{r_{11}(8)}{r_{21}(8)}$, where $r(8)$ denotes the received signal at $t = 8$.



Ans:

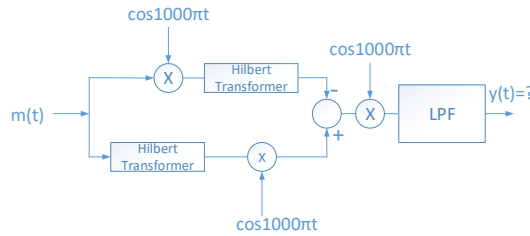
$$\infty$$

- Q7. In an AWGN channel $y(t) = x(t) + n(t)$, the signal power is fixed at 20dBm and the noise PSD $N_0/2 = -120$ dBW/Hz. Find the channel capacity (in bps) if the channel bandwidth becomes unbounded i.e., $B \rightarrow \infty$.

Ans:

$$0.72 \times 10^{11} [2 \text{ Marks}] \text{ or } 1.44 \times 10^{11} [1 \text{ Mark}]$$

- Q8. Find $y(t)$ in terms of $m(t)$ as shown in the system diagram below. The frequency response of both message signal $m(t)$ and LPF is $\Pi(f/200)$.



Ans:

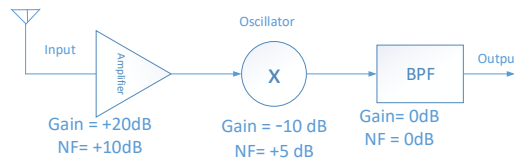
$$\frac{m_h(t)}{2}$$

- Q9. A voice signal has a frequency range 300 Hz- 3300 Hz. It is sampled at the Nyquist rate. A 4-bits/sample PCM encoding ensures transmission at the channel capacity. For data transmission, M-ary transmission using Nyquist pulses with a roll-off factor 0.5 is used. Find the value M if the channel bandwidth is 3000 Hz.

Ans:

$$64$$

- Q10. A part of super-hetrodyne receiver is shown in the figure below. The noise PSD (two sided) at the input of amplifier is $N_0/2 = -100$ dBm/Hz. Find the noise power (in dBm) at the output of band-pass filter. The bandwidth of BPF is 50 Hz.



Ans:

$$-59.9906$$

Solution: Detailed

Q1. $\Delta_f = \max \frac{1}{2\pi} \frac{d}{dt} 12\pi m(t) = 6 \max \frac{d}{dt} m(t) = 6 \times (2/6) = 2.$

Ans:

2

Q2. Combine three alphabets at one time, and coding as 012 using Huffman code gives codes as 0, 10, 11, 12, 20, 21, 22. Average length = $0.22 + 2 \times (0.18 + 0.17 + 0.15 + 0.13 + 0.10 + 0.05) = 1.78$

Ans:

1.78

Q3. Codes $c(x) = d(x)g(x)$: When $d = 00$, code = 0000, when $d = 01$, code = 0101, when $d = 10$, code = 1010, when $d = 11$, code = 1111. $d_{min} = 2$, error detection $d_{min} = t + 1$ gives $t = 1$.

Ans:

1

Q4. Write down two types of single-user spread spectrum techniques.

Ans:

Frequency Hopping and Direct Sequence

Q5. Signal points are at distance 2 and $2\sqrt{2}$ units from the origin. Average power $\frac{1}{8}[4 \times 2^2 + 4 \times (2\sqrt{2})^2] = 6$. Average bit power = $6/3=2$.

Ans:

Symbol power= 6 or Bit power= 2

Q6. Output signal $r_{21}(8) = f_2(t) * s_1(t) = 0$ at $t = 8$. Thus the ratio will be infinity.

Ans:

∞

Q7. Capacity $C = B \log_2(1 + S/N_0B)$. When limit $B \rightarrow \infty$, $C = 1.44S/N_0$. Converting in linear scales, gives the result

Ans:

$$1.44 \frac{10^2 \times 10^{-3}}{2 \times 10^{-12}} = 0.72 \times 10^{11}.$$

Q8. Writing the output signals and noting that LPF passes only the signal within the BW of $m(t)$:

Ans:

$$\frac{m_h(t)}{2}$$

Q9. Nyquist rate 6600 samples/second. PCM output= capacity= 6000 samples/second $\times$ 4 bits/sample = 26400 bps. Using BW = 3000 and $\alpha = 0.5$, $R_s = \frac{2B}{1+\alpha} = 4000$ symbols/ second. Each symbol should carry 6.6 bits. Hence $M = 2^6 = 64$. **We can not take $b = 7$ or $M = 128$ since it will exceed the channel capacity.**

Ans:

64

Q10. Resultant NF combined with amplifier and oscillator (since BPF is noiseless) $= 10 + \frac{10^{0.5} - 1}{100} = 10.0216$. Output power (product of noise PSD with BW of BPF and gains in linear scale 100 (amplifier) and 10 (oscillator) and NF (10.0216)) $= 10^{-10} \times 2B \times 10.0216 \times 100 \times 1/10 = 1.0022 \times 10^{-6} mW = -59.9906 \text{ dBm}$.

Ans:

-59.9906

Birla Institute of Technology & Science, Pilani
1st Semester 2019-2020
Comprehensive Exam-Part B (Open Book)

Course No: EEE F311

Course Title: Communication Systems

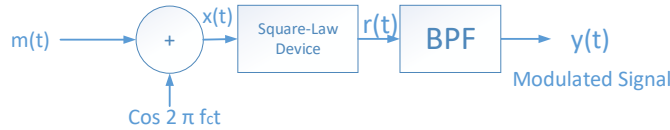
Maximum Marks: 50

Duration: 120 Minutes

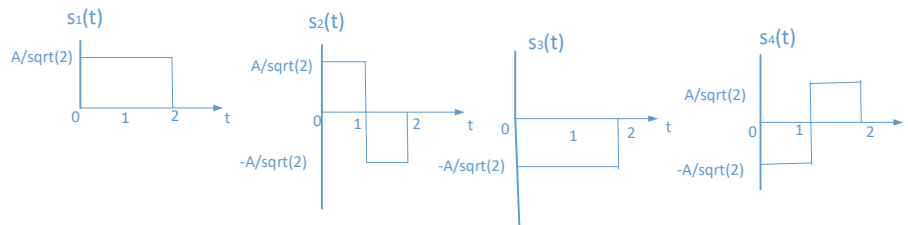
Date: 09-12-2019

Write down assumptions and definitions wherever there is an ambiguity.
Please write the final answer and important steps in rectangular boxes.

- Q1. An analog passband transmission is shown in the figure below. The bandpass filter has a bandwidth $2B$ centered at f_c . The message signal $m(t)$ is real-valued signal with amplitude varying between $-A_m$ to A_m , and has an average power P_m . The bandwidth of $m(t)$ is B . The square-law device obeys $r(t) = 2x(t) + x^2(t)$. **[8 Marks]**.

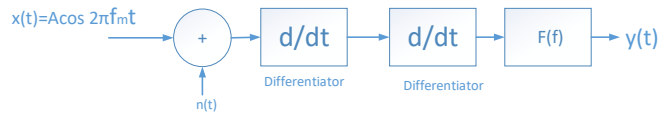


- (a) Find the average transmitted power of the modulated signal $y(t)$ in terms of P_m .
(b) Find the modulation index of the modulated signal $y(t)$.
(c) Draw a system block diagram for coherent demodulation.
(d) What minimum value A_m will ensure non-coherent demodulation?
- Q2. An M-ary PAM communication system transmits data at a rate 24000 bps. The communication channel passes signals with frequencies from 600 Hz to 6600 Hz. **[6 Marks]**.
- (a) If $M = 16$, find the pulse (in time-domain) satisfying the Nyquist criteria for zero-ISI.
(b) If $M = 4$, find the pulse (in time-domain) satisfying the Nyquist criteria for zero-ISI.
(c) If $M = 8$, find BER (in numerical value) if $E_b/N_0 = 3.5$.
- Q3. A communication system uses 4-ary signaling where the signal set is as shown in the figure below. For parts (a) to (e), assume that there is no effect of channel. **[12 Marks]**.

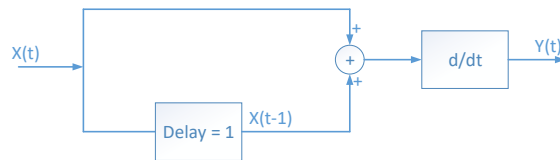


- (a) Find average energy of the transmitted signals.
(b) Sketch matched filter $f_3(t)$ corresponding to transmitted signal $s_3(t)$.

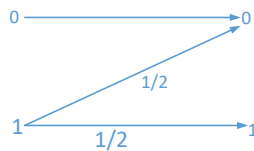
- (c) Find the signal at the output of the matched filter $f_3(t)$ if $s_1(t)$ is transmitted.
- (d) Find minimum number of basis signals and express the signals $s_1(t)$, $s_2(t)$, $s_3(t)$, and $s_4(t)$ using the basis signals.
- (e) Find the BER (in terms of signal amplitude A and Q-function) under AWGN of PSD $N_0/2$.
- (f) Sketch the matched filters (using basis signals) if there is a distortionless channel $h(t) = \delta(t - 5)$.
- Q4. A message signal $x(t) = A \cos(2\pi f_m t)$ is received under additive white Gaussian noise having two-sided PSD of $N_0/2$ W/Hz. The receiver is realized by two differentiator followed by a unity gain LPF filter of bandwidth B i.e., $F(f) = 1, -B \leq f \leq B$. Note that $f_m < B$ and that $y(t)$ contains both the signal and noise components. [6 Marks].



- (a) Find the signal power at the output of LPF.
- (b) Find the average noise power at the output of LPF.
- Q5. $X(t)$ is a random process with auto-correlation function $R_x(\tau) = e^{-|\tau|}$. The process is applied to a linear system where $X(t)$ is input and $Y(t)$ is output as shown in the figure below. [4 Marks].
- (a) Find the transfer function $H(f)$ of the system.
- (b) Find $S_y(1)$ i.e., the PSD of the random process $Y(t)$ for $f = 1$.



- Q6. Consider a linear systematic block encoder where three information bits $\mathbf{d} = (d_1, d_2, d_3)$ is complemented with three parity check bits as $c_1 = d_2 \oplus d_3$, $c_2 = d_1 \oplus d_3$, and $c_3 = d_1 \oplus d_2$. [6 Marks].
- (a) Find the code word for the data 101.
- (b) Construct the parity check matrix $\mathbf{Q}$ such that $\mathbf{GQ}^T = 0$.
- (c) Decode the received word 100001.
- Q7. Determine the capacity of the channel described by the channel matrix as shown in the figure below. Note that the input symbols are not equiprobable. [4 Marks].



- Q8. Consider a power-constrained additive noise channel $Y = X + Z$, where X and Y are the input and the output signals and Z is the noise. Let S be the power of signal X and N be the power of noise Z . Take channel bandwidth $B = 1$ Hz. Find the channel capacity if the additive noise Z is exponentially distributed i.e. $f_Z(z) = \lambda e^{-\lambda z}, z \geq 0$. [4 Marks].

Solution

Q1. After calculations and passing signal at centered frequency f_c : $y(t) = 2 \cos 2\pi f_c t + 2m(t) \cos 2\pi f_c t$

- (a) Average power: $\frac{4}{2} + 4\frac{P_m}{2} = 2 + 2P_m$
- (b) Modulation index = A_m
- (c) Multiplication with $\cos 2\pi f_c t$ then LPF yield $1 + m(t)$, then pass through DC blocker to get $m(t)$. Thus the block diagram should consists of multiplier with $\cos 2\pi f_c t$, followed by LPF, then a DC blocker.
- (d) $|2 + 2m(t)| \geq 0$. Using $(2 + 2m(t)) \cos 2\pi f_c t$ or the modulation index, the minimum value of $A_m = 0$.

Q2. $R_b = 24000$ bps, $B = 6000$.

- (a) If $M = 16$, $R_s = 6000$ sps. Using $R_s = \frac{2B}{1+\alpha}$, we get $\alpha = 1$. This gives raised cosine pulse $p(t) = R_s \frac{\cos \pi R_s t}{1-4R_s^2 t^2} \text{sinc}(\pi R_s t)$, where $R_s = 6000$.
- (b) If $M = 4$, $R_s = 12000$ sps. Using $R_s = \frac{2B}{1+\alpha}$, we get $\alpha = 0$. This gives sinc pulse $p(t) = \text{sinc}(\pi R_s t)$, where $R_s = 12000$.
- (c) If $M = 8$, $\text{SER} = 2 \frac{M-1}{M} Q[\sqrt{\frac{6 \log_2 M}{M^2-1} (\frac{E_b}{N_0})}]$. This gives $\frac{7}{4} Q(1)$ or $\frac{7}{4} \times 0.1587$ or 0.2777 [1 Mark]. $\text{BER} = 0.2777 / (\log_2 M) = 0.2777 / 3 = 0.0926$ [1 Mark].

Q3. (a) $(A^2 + A^2 + A^2 + A^2)/4 = A^2$.

(b) Sketch of $f_3(t)$ is exact same as $s_3(t)$.

(c) Output at the matched filter. A sketch will show a triangular pulse of amplitude $-A^2$ at $t = 2$.

$$y(t) = \begin{cases} 0, & t < 0 \\ -\frac{A^2}{2}t, & 0 \leq t \leq 2 \\ \frac{A^2}{2}(t-4), & 2 \leq t \leq 4 \\ 0, & t > 4 \end{cases}$$

(d) Number of basis signals 2: $\phi_1(t)$: rectangular pulse of height 1 and width 1: time-axis: 0 : 1, $\phi_2(t)$: rectangular pulse of height 1 and width 1 time-axis: 1 : 2. [1 Mark] $s_1(t) = \frac{A}{\sqrt{2}}\phi_1(t) + \frac{A}{\sqrt{2}}\phi_2(t)$, $s_2(t) = \frac{A}{\sqrt{2}}\phi_1(t) - \frac{A}{\sqrt{2}}\phi_2(t)$, $s_3(t) = -\frac{A}{\sqrt{2}}\phi_1(t) - \frac{A}{\sqrt{2}}\phi_2(t)$, $s_4(t) = -\frac{A}{\sqrt{2}}\phi_1(t) + \frac{A}{\sqrt{2}}\phi_2(t)$. [1 Mark] **Note that there are other basis signals possible. However, there must be 2 basis signals which must be of unit energy and orthogonal to each other.**

(e) The above basis signals can be used to represent signals as a 4-QPSK/4-QAM constellation point. Error probability of 4-QAM/4-QPSK: $2Q\left(\sqrt{\frac{A^2}{N_0}}\right) - Q^2\left(\sqrt{\frac{A^2}{N_0}}\right)$.

(f) Matched filter 1: Shift $\phi_1(t)$ right 5 units. Matched filter 1: Shift $\phi_2(t)$ right 5 units.

Q4. (a) Double derivative on signal: $-(2\pi f_m)^2 \cos 2\pi f_m t$. Signal power = $8A^2\pi^4 f_m^4$.

(b) Noise PSD at the output due to two transfer functions of differentiation: $N_0/2(2\pi f)^4$. Noise power $(N_0/2)16\pi^4 \int_{-B}^B f^4 df = \frac{16}{5}N_0\pi^4 B^5$.

Q5. (a) $H(f) = j2\pi f + j2\pi f e^{-j2\pi f}$

(b) $S_x(f) = \frac{2}{1+4\pi^2 f^2}$, $S_y(f) = S_x(f)|H(f)|^2$. $S_y(1) = \frac{32\pi^2}{1+4\pi^2} = 7.8024$.

Q6. (a) The code word: 101101.

(b) Given:

$$\mathbf{Q}^T = \begin{bmatrix} 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

(c) Decoded word 100.

Q7. Capacity 0.322 bits/ symbol.

Let $P(X = 1) = p$, $P(X = 0) = 1 - p$. $H(Y) = -(1 - p + p/2) \log_2(1 - p + p/2) - (p/2) \log_2(p/2) = -(1 - p/2) \log_2(1 - p/2) - (p/2) \log_2 p/2$.

$H(Y/X) = (-p/2) \log_2 1/2 - (p/2) \log_2(1/2)$.

$I(X; Y) = H(Y) - H(Y/X)$.

To find p , we differentiate with $I(X; Y)$ and equate to zero. This gives $p = 2/5$. Using this, capacity is 0.322 bits/ symbol.

Q8. Capacity $C = \log_2(2\pi eS + N) - 2H(n)$ bps. Here $H(n)$ is the differential entropy of exponential random variable. $H(n) = -\int_0^\infty \lambda e^{-\lambda z} \log_2(\lambda e^{-\lambda z}) dz = \log_2 \frac{e}{\lambda}$. Hence, $C = \log_2(2\pi eS + N) - 2 \log_2 \frac{e}{\lambda}$ bps. Capacity in bits per symbol will be $C = \frac{1}{2} \log_2(2\pi eS + N) - \log_2 \frac{e}{\lambda}$.
